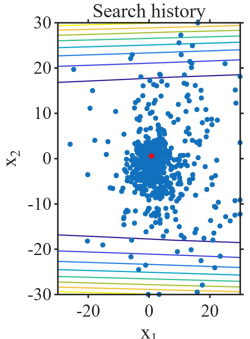
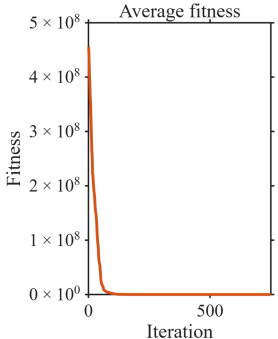
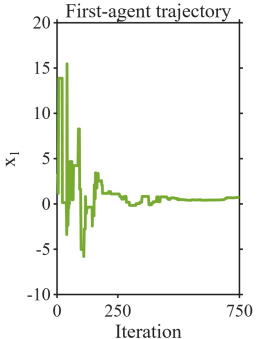
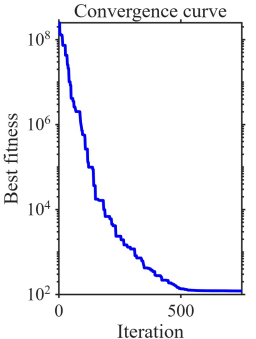

F5
×10^10
15
10
5
Fitness
200
0
-200
x2
-200
0
200
x1


Search history
30
20
10
0
-10
-20
-30
x2
-20
0
20
x1


Average fitness
5 × 10^8
4 × 10^8
3 × 10^8
2 × 10^8
1 × 10^8
0 × 10^0
Fitness
0
500
Iteration


First-agent trajectory
20
15
10
5
0
-5
-10
x1
0
250
750
Iteration


Convergence curve
10^8
10^6
10^4
10^2
Best fitness
0
500
Iteration